GGG-B-671F
February 11, 1976
SUPERSEDING
Fed. Spec. GGG-B-671E
January 11, 1962

FEDERAL SPECIFICATION

BRACE, BIT, RATCHET; BRACE, BIT, CORNER; AND EXTENSIONS, BIT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers ratchet bit braces, corner bit braces, and bit extensions generally used in woodworking for drilling and boring, with round, square or tapered-shank bits.

1.1.1 Federal specification coverage. Federal specifications do not include all types and classes of the commodity which are available, but cover only those which are generally purchased by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Ratchet bit braces, corner bit braces, and bit extensions shall be of the following types, classes, and styles as specified (see 6.1).

Type I - Ratchet bit brace.
  Class 2 - Box ratchet.
    Style A - Ball bearing chuck.
    Style B - Plain chuck.
Type II - Corner bit brace.
Type III - Close-quarter, ratchet bit brace.
Type IV - Bit extension.
    Style A - Square socket.
    Style B - Split collet or jaw.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standard:

Fed. Std. No. 358 - Sampling Procedures.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive and not restrictive and are included for the convenience of identification and are not intended to preclude the purchase of tools which are otherwise in accordance with this specification.

3.2 Material. The material from which the braces and bit extensions are constructed shall be of a quality to produce tools conforming to the requirements as hereinafter specified. The strength and other physical properties shall be adequate throughout to withstand the maximum strain to which the tools may be subjected under service conditions.

3.3 Coating. Unless otherwise specified (see 6.1), all exposed metal surfaces of the braces and bit extensions shall have one of the following coatings for protection from rust and corrosion:

(a) Bright chromium plate.
(b) Bright nickel plate.

3.4 Finish. All exposed surfaces of the wood-center handgrip and head shall be smoothly finished with a protective and durable coating of varnish or lacquer to prevent the absorption of moisture.

3.5 Type I, ratchet bit brace. Type I ratchet bit brace shall consist essentially of a chuck, ratchet mechanism, sweep arm, center handgrip, and antifriction bearing head. Type I brace shall conform to the concentricity of 4.4.2.4 and to the dimensions shown in table I for the sweep specified (see 6.1).

3.5.1 Chuck. The chuck shall be of steel consisting of a sleeve, sleeve assembly, or shell; spindle; and two universal jaws. The chuck shall be capable of securely gripping and holding the shanks of bits by hand tightening only. The chuck shall also withstand the test specified in 4.4.2.1.

3.5.1.1 Sleeve, sleeve assembly, or shell. The sleeve, sleeve assembly, or shell shall be steel. The outside diameter of the sleeve shall be knurled in such a manner as to provide a secure handgrip. The largest inside diameter of the sleeve, sleeve assembly, or shell shall be threaded for attaching the sleeve to the spindle. The interior surface at the point end of the sleeve shall be tapered to fit the tapered portion of the chuck jaws. The tapered surface of the sleeve shall be of sufficient hardness to prevent the jaws from cutting into the body of the sleeve.

3.5.1.2 Spindle. The spindle shall be of steel. The outside diameter of the spindle shall be threaded to properly accommodate the internal thread of the chuck sleeve, sleeve assembly, or shell. The threads shall be concentric with the spindle within 0.020-inch-total indicator reading throughout the entire length of the thread. The spindle shall be provided with a slot of sufficient depth to receive the chuck jaws. The back or socket end of the spindle shall be provided with an adjustable nut and cotter pin or other suitable means for securing the spindle in the housing.

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TABLE I. Type I, ratchet bit brace

Sweep [1]	Sleeve				Head				Center handgrip			
	Overall length		Outside diameter		Diameter		Thickness		Length		Diameter	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inches	Inches	Inches	Inches	Inches
8 10 12 14	1-3/4	3-1/2	1-3/8	1-3/4	2-3/4	3-1/4	7/8	1-1/4	2-5/8	3-1/2	1-1/8	1-3/8

[1] The sweep shall be the diameter of the circle traversed by the center handgrip.

3.5.1.3 Universal jaws. The universal jaws shall be of steel or sintered steel and copper or other suitable alloy. The steel jaws shall show a Rockwell hardness of not less than 50 nor more than 58 on the "C" scale or be casehardened to a depth of not less than 0.015 inch showing a hardness value of not less than 84.5 nor more than 89.0 on the Rockwell "15N" scale (superficial) (see 4.4.1). The sintered steel and copper jaws shall show a hardness value of not less than 66 nor more than 79 on the Rockwell "15N" scale (see 4.4.1). The two jaws shall be provided with a steel wire spring or other suitable means to exert sufficient tension to cause the jaws to automatically spread as the chuck is opened. The chuck jaws shall withstand the test specified in 4.4.2.1. The jaws shall be designed to grip and hold drills and auger bits having all of the following shanks:

(a) Tapered square shank from 3/16 inch to 1/4 inch on the small end and from 5/16 inch to 7/16 inch on large end.
(b) Straight round shanks of diameters from 1/8 inch to 1/2 inch.

3.5.2 The braces shall withstand the compression test specified in 4.4.2.2.

3.5.3 Ratchet mechanism. The ratchet mechanism shall be manually operated with means provided for permitting the selection of directional rotation of the chuck. The action of the chuck relative to the position of the ratchet mechanism shall be as follows (see 4.4.2.3):

(a) When the ratchet is positioned for right-hand rotation, the chuck shall rotate in a right-hand direction with a right-hand rotation of the sweep arm. No movement of the chuck shall take place with a left-hand rotation of the sweep arm.
(b) When the ratchet is positioned for left-hand rotation, the chuck shall rotate in a left-hand direction with a left-hand rotation of the sweep arm. No movement of the chuck shall take place with a right-hand rotation of the sweep arm.
(c) The ratchet mechanism shall be provided with a lock position for permitting a direct rotational drive of the chuck with either the right or left-hand rotation of the sweep arm.

3.5.4 Sweep arm. The sweep arm shall be one solid piece of steel. The arm shall have a smooth surface free from any imperfections which would interfere with the serviceability.

3.5.5 Center handgrip. The handgrip shall be well proportioned and of a size and shape to facilitate a comfortable handgrip.

3.5.6 Antifriction bearing head. The head shall be firmly secured to the metal quill disc. The sweep arm at the head end shall be provided with a steel bearing cup and ball bearings. A hardened thrust washer shall be placed between the quill end and bearing cup to receive the thrust load applied to the head. The head shall be provided with means for lubricating the bearings and bearing surfaces. The complete assembly of the antifriction bearing head shall be attached to the sweep arm in such a manner that the end of the sweep arm will not pass through the center of the head.

3.5.7 Ratchet housing or socket. The ratchet housing or socket shall be of steel, malleable iron, aluminum alloy or bronze. The housing shall be designed to completely cover all working parts of the ratchet mechanism to provide an adequate seal against dirt and grit. The housing shall be permanently secured to the sweep arm, in a manner not to loosen during the life of the tool.

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3.5.8 Class 2, box ratchet. Class 2 box ratchet shall consist essentially of ratchet teeth, two pawls of steel, malleable iron or sintered steel and copper or other suitable alloy, and ratchet control. The ratchet teeth and pawls shall be of sufficient hardness to prevent excessive wear. The ratchet control shall be positive in action and shall not move or become loose when positioned. The ratchet shall be similar to figure 1.

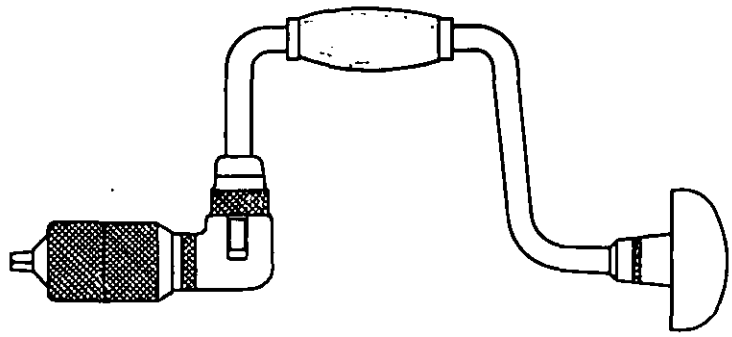


FIGURE 1. Type I, class 2, ratchet bit brace, box ratchet

3.5.9 Chuck. The chuck shall conform to 3.5.9.1 or 3.5.9.2.

3.5.9.1 Ball-bearing chuck. The ball bearing chuck shall conform to 3.5.1, except the chuck shall be provided with ball bearings. The nose end of the chuck sleeve or shell shall have a ring or collar of sufficient hardness to provide a bearing race. The ring or collar shall be attached to the sleeve to completely enclose and protect the ball bearings from dirt and grit. Means shall be provided for lubricating the ball bearings.

3.5.9.2 Plain chuck. Plain chuck shall be in conformance with 3.5.1.

3.5.10 Center handgrip. Center handgrip shall conform to 3.5.10.1 or 3.5.10.2.

3.5.10.1 Plastic center handgrip and head. Plastic center handgrip and head shall be constructed from one piece and shall be free from cracks, chipped edges, uneven surfaces, blisters, warps, dents, heat marks, or any other defects that may affect their serviceability. *The hand grip and head may be plain or colored at the option of the manufacturer.* The center handgrip and head shall conform to 3.5.5 and 3.5.6.

3.5.10.2 Hardwood center handgrip and head. Hardwood-center handgrip and head shall be of clear straight grained cocobolo, rosewood, birch, goncalo, or lignum vitae. The handgrip and head shall be free from knots, checks, holes, or any other defects. The center handgrip shall be protected and reinforced at each end by metal collars or a ring to prevent splitting and chipping of the grip. The handgrip shall be held in place by the collars or rings, swaged, or secured to the sweep arm at each end of the grip in a manner to prevent their becoming loose and also allow the handgrip to rotate freely on the sweep arm.

3.6 Type II, corner bit brace. Type II corner bit brace nonratchet shall consist essentially of a chuck, two bevel gears, gear housing, frame rod, sweep arm, two handgrips, and head. Type II braces shall withstand the compression test specified in 4.4.3.1. The corner brace shall conform to the dimensions shown in table II and shall be similar to figure 2.

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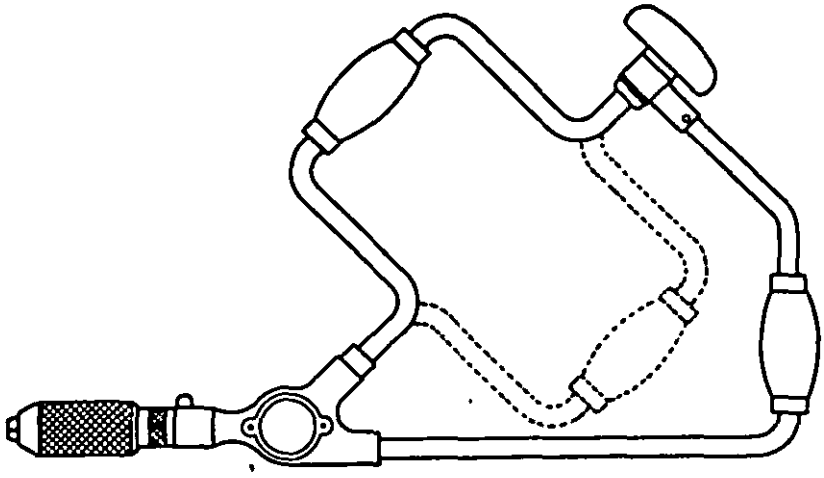


FIGURE 2. Type II, corner bit brace, *nonratchet*

TABLE II. Type II, corner bit brace, nonratchet

Sweep [1]	Sleeve				Head				Center handgrip			
	Overall length		Outside diameter		Diameter		Thickness		Length		Diameter	
±1.000	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inches	Inches	Inches	Inches	Inches
9	2-1/2	3-1/2	1-3/8	1-3/4	2-3/4	3-1/4	7/8	1-1/4	2-5/8	3-1/4	1-1/8	1-3/8

[1] The sweep shall be the diameter of the circle traversed by the center handgrip.

3.6.1 Chuck. The chuck shall be in conformance with 3.5.1 and shall withstand the tests specified in 4.4.3.2 and 4.4.3.3.

3.6.2 Two bevel gears. The two bevel gears shall be of steel properly hardened to a sufficient depth to prevent excessive wear. The gears shall have machine-cut bevel teeth free from any imperfections, burrs, or defects of any nature. The tooth contact on the face width of the gears shall be not less than 3/16 inch with a minimum backlash of not more than 0.010 inch. The gears shall roll freely when assembled.

3.6.3 Gear housing. The gear housing shall be of either steel, bronze, or malleable iron. The housing shall be designed to completely enclose the two bevel gears and provide an adequate seal against dirt and grit. The housing shall be permanently secured to the frame rod. The bearing for the sweep arm shall have a smooth machine finish forming an approximate angle of 45 degrees from the longitudinal axis of the frame rod.

3.6.4 Frame rod. The frame rod shall be one piece of solid steel of a strong and durable construction. The frame rod shall be secured to the head of the sweep arm in such a manner as not to become loose. The rod shall be provided with a handgrip for applying pressure to the chuck.

3.6.5 Sweep arm. The sweep arm shall be in conformance with 3.5.4.

3.6.6 Handgrips and head. The handgrips and head shall be in conformance with 3.5.5, 3.5.6, and 3.5.10.2.

3.7 Type III, close-quarter ratchet bit brace. Type III, close-quarter ratchet bit brace shall consist essentially of a chuck, ratchet mechanism, handgrip and head unit. Type III ratchet bit brace shall conform to the dimensions shown in table III, and shall be similar to figure 3.

3.7.1 Chuck. The chuck shall be in accordance with 3.5.1 and shall withstand the test specified in 4.4.2.1.

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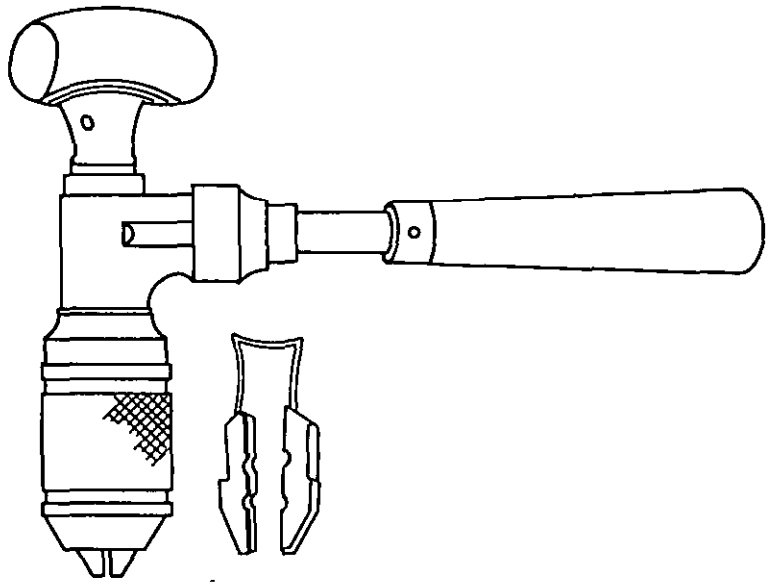


FIGURE 3. Type III, close-quarter ratchet bit brace

TABLE III.—*Type III, close-quarter ratchet bit brace*

Height, maximum	Sleeve				Head				Handgrip				Torque arm	
	Overall length		Outside diameter		Diameter		Thickness		Length		Diameter		Length[1]	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
7½	2½	8½	1⅝	1⅝	2⅝	3⅛	⅛	1⅛	4½	9	⅛	1⅝	7½	12

[1]Length as measured from the center line of spindle to the end of the handle.

3.7.2 Sleeve, sleeve assembly, or shell. The sleeve, sleeve assembly, or shell shall be in conformance with 3.5.1.1.

3.7.3 Universal jaws. The universal jaws shall be in conformance with 3.5.1.3.

3.7.4 Ratchet mechanism. The ratchet mechanism shall be in conformance with 3.5.3 and shall withstand the test specified in 4.4.4.2.

3.7.4.1 Ratchet housing or socket. The ratchet housing or socket shall be of either steel, malleable iron, or bronze. The housing shall be designed to completely cover all working parts of the ratchet mechanism and provide an adequate seal against dirt and grit.

3.7.4.2 Box ratchet. Type III, box ratchet shall consist essentially of ratchet teeth, two pawls of steel or malleable iron, and ratchet control. The ratchet teeth and pawls shall be of sufficient hardness to prevent excessive wear. The ratchet control shall be positive in action and shall not move or become loose when positioned. The ratchet shall be similar to figure 4.

3.7.5 Handgrip and head. The handgrip and head shall be constructed of clear straight-grained cocobolo, rosewood, birch, goncalo, or lignum vitae. The handgrip and head shall be free from knots, checks, holes, or other defects. The handgrip shall be protected and reinforced at one end by a metal collar or a ferrule to prevent splitting and chipping of the handgrip. The handgrip shall be secured in place by a metal rivet or pin through at least one side of the collar or ferrule and into the torque arm or other means shall be provided to retain the handgrip in place under maximum usage. The head shall be slabbed on one side to enable the user to bore holes close to perpendicular surfaces.

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3.7.6 Head unit. The head unit shall consist essentially of a wooden head and a metal sleeve section. The wooden head shall be secured to the metal sleeve section in such a manner that it will not become detached or loosen under maximum usage. The metal sleeve section of the head unit shall rotate freely on the spindle and shall not become detached from the spindle under maximum usage. A means shall be provided for lubricating the bearing surfaces. The head unit shall be attached to the spindle in such a manner that the end of the spindle will not pass through the head.

3.7.7 Spindle. The spindle shall be of steel. The outside diameter of the spindle shall be threaded to properly accommodate the internal thread of the chuck sleeve, sleeve assembly or shell. The threads shall be concentric with the spindle within 0.020-inch-total indicator reading, throughout the entire length of the thread. The spindle shall by provided with a slot of sufficient depth to receive the chuck jaws. The spindle shall be provided with suitable means for securing the spindle in the housing. The head end of the spindle shall be provided with a positive means for retaining the head unit.

3.8 Type IV, bit extension. Bit extensions shall be designed for use in bit braces, and shall be capable of securely gripping squared taper-shank bits and withstand the test specified in 4.4.5.1. The largest diameter of the extension shall be capable of following through an 11/16-inch-diameter hole and shall be designed to hold bits up to and including 3/4 inch in diameter. The bit extensions shall be furnished in the following nominal lengths: 15, 18, 21, 24, and 30 inches (see 6.1).



FIGURE 4. Type IV, style A, bit extension, square socket

3.8.1 Style A, square socket. Style A square socket shall consist essentially of a rod, socket, sleeve, and nut. Means shall be provided to prevent the bit from slipping in the socket or being left in the material drilled. Style A square socket shall be similar to figure 4.

3.8.1.1 Rod. The rod shall be of steel. One end of the rod shall have a squared taper shank that can be gripped securely in the chuck of a bit brace (see 3.5.1.3). The opposite end of the rod shall provide means for attaching the socket.

3.8.1.2 Socket. The socket shall be of steel of sufficient hardness to prevent excessive wear. The end of the socket shall have a squared, tapered hole to receive the squared taper shanks of bits. The socket shall be securely attached to the rod with a connection to provide a positive drive.

3.8.1.3 Sleeve. The sleeve shall be of steel with an opening at the front end to permit a square taper shank bit to engage the socket. The opposite end of the sleeve shall be threaded internally with a class 2 or 3 thread to facilitate tightening the bits in the chuck by means of an external threaded nut. Means shall be provided to prevent the sleeve from becoming detached from the rod.

3.8.1.4 Nut. The nut shall be of steel and free to rotate on the rod. The outside diameter of the nut shall have a class 2 or 3 thread to engage the internal thread of the sleeve. The thread shall be of a sufficient length that the draw of the sleeve will rigidly hold the bits in the socket. The nut shall not bind or jam when fully engaged in the sleeve and shall be easily released without the aid of tools. The outside diameter at the back portion of the nut shall be knurled for tightening the nut by hand.

3.8.2 Style B, split collet or jaw. Style B split collet or jaw shall be in conformance with 3.8.1 except the extension shall be provided with a split collet or jaw, and shall be similar to figure 5.

3.8.2.1 Split collet or jaw. The collet or jaw shall be of steel and split in two equal halves. The two halves of the collet or jaw shall show a Rockwell hardness value of not less than 40 nor more than 53 on the "C" scale (see 4.4.1). The outside diameter of the collet or jaw shall be tapered from the front end to the opposite end that is attached to the rod. The inside diameter shall have tapered grooves or other suitable means for securely holding squared taper shank bits. The collet or jaw shall be attached to the rod so as to allow the two halves of the collet or jaw to spread when inserting the auger bit.

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3.9 Marking. Braces and bit extensions shall be marked in a permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.10 Workmanship. The braces and bit extensions shall be free from any imperfections which may affect their appearance, durability, or serviceability.



FIGURE 5. Type IV, style B, bit extension, split collet or jaw

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with FED-STD-358.

4.3 Testing. Each sample unit shall be tested in accordance with 4.3.1 and 4.4.1 thru 4.4.5 as applicable.

4.3.1 Hardness. Hardness tests shall be conducted in accordance with ASTM E-18. When grinding is necessary to prepare the test surface, the amount of material removed must not exceed 0.007 inches on the surface contacted by the indentor.

4.4 Tests.

4.4.1 Hardness test of jaws. Jaws of each sample brace and bit extension hardened and tempered shall be tested on a Rockwell "C" scale or, when constructed of sintered steel and copper, on a Rockwell "15N" scale (superficial) to determine conformance with 3.5.1.3 and 3.8.2.1.

4.4.2 Type I, ratchet bit brace.

4.4.2.1 Chuck test. The brace shall be gripped in a bench vise with the chuck in a horizontal plane and 1/2-inch-diameter straight-shank drill tightened in the chuck under a torque to 100 inch-pounds by revolving the chuck sleeve. The brace shall then be removed from the vise, and one hole shall be drilled through one inch of hardwood such as ash, oak, or mulberry. The 1/2-inch straight-shank drill shall be removed, and the test repeated using 1/2-inch auger bit with tapered square shank. The drills shall not slip, turn or distort the chuck jaws under this test. The 100 inch-pounds may be obtained by using a torque wrench or an ordinary pipe wrench, spring balance, and a pry bar. The pipe wrench, when used, shall grip the chuck sleeve with the load applied at right angles to the predetermined point on the handle end of the wrench. The magnitude of the torque will depend on the distance the load is applied from the longitudinal axis of the chuck sleeve. Ten pounds applied at 10 inches, 50 pounds applied at 2 inches, etc., from the axis will produce a torque of 100 inch-pounds, the inch-pounds being the product of the leverage in inches multiplied by the load in pounds (see 3.5.1).

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4.4.2.2 Compression test. (1) Each of the sample braces shall be tested in the following manner (see 3.5.2). (2) The sample brace shall be placed in a vertical position and a load of 20 pounds pressure shall be applied to the top face of the head in the direction of the longitudinal axis of the chuck. (3) The measurement of the overall length from the outermost end of the chuck to the top face of the sweep head shall be made. (4) The contraction in this dimension shall not exceed 9/16 inch when the load is 200 pounds (total) for the 8-inch and 10-inch sweep, and when the load is 150 pounds (total) for the 12-inch and 1/4-inch sweep. A permanent set of not more than 1/16 inch shall be allowed when the load is decreased to 20 pounds.

4.4.2.3 Torque and ratchet mechanism test. A squared shank bit or piece of steel of a suitable size and shape shall be gripped in the chuck jaw of the brace, with the outer end of the bit or piece of steel clamped in a bench vise. The head of the brace shall be supported and the sweep arm placed in a horizontal plane. A vertical load of 80 pounds shall be applied at or near the center of the handgrip for each ratchet position. No failure, perceptible deformation, or damage to the ratchet or ratchet brace shall occur in either of the ratchet positions (see 3.5.3).

4.4.2.4 Concentricity of brace. The concentricity for type I ratchet braces shall be tested as shown on figure 6. The fixture employed to determine the concentricity shall consist of a table or plate, block and mandrel. The block shall be secured to the plate and provided with a hole to receive the mandrel. The mandrel shall be the same diameter as the sweep arm and allowed to rotate in the block. The chuck of the brace shall be tightened on the mandrel as illustrated at point 1 with the sweep arm in an upright position. The brace shall be tightened on the mandrel so that the indicator height reading for the distance from the table or plate to point 1 shall be approximately the same distance at point 2. With the indicator at point 2, the sweep arm shall be revolved 360 degrees and the concentricity of the brace (see 3.5) shall be within the following applicable total indicator reading: 0.250-inch for the 8-inch, 0.375-inch for the 10-inch, 0.500-inch for the 12-inch, and 0.625 for the 1/4-inch sweep (size brace).

4.4.2.5 Plastic head and handgrip.

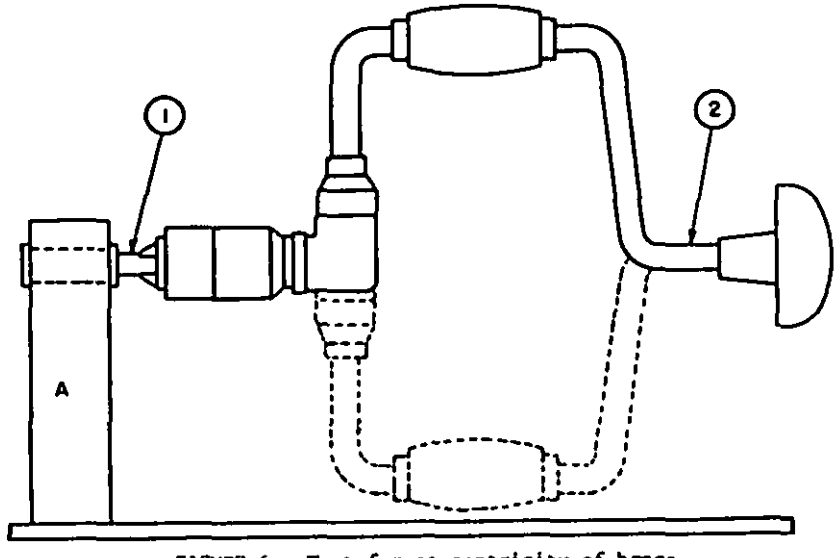


FIGURE 6. Test for concentricity of brace

4.4.3 Type II, corner brace.

4.4.3.1 Compression test. Each sample corner brace shall be tested in conformance with 4.4.2.2 except the load shall be applied on the top face of the frame rod.

4.4.3.2 Chuck test. The chuck test shall be conducted in the same manner as specified in 4.4.2.1.

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4.4.3.3 Concentricity of chuck. The frame rod between the handgrip and gear housing shall be gripped in a bench vise with the chuck in a horizontal plane. A test rod of any diameter within the nominal capacity range of the chuck shall be properly inserted between the jaws, and the chuck or chuck sleeve tightened by hand. The test rod shall extend 4 inches from the face of the jaws when inserted to the full depth of the chuck. When the chuck is revolved 360 degrees the test rod shall run concentric within 0.060-inch-total indicator reading throughout its entire length (see 3.6.1).

4.4.4 Type III, close-quarter, ratchet, bit brace.

4.4.4.1 Chuck test. The chuck test shall be conducted in the same manner as specified in 4.4.2.1.

4.4.4.2 Torque and ratchet mechanism test. The torque and ratchet mechanism test shall be conducted in the same manner as specified in 4.4.2.3.

4.4.5 Type IV, bit extension.

4.4.5.1 Torque test. Each of the sample bit extensions shall be tested in the following manner. A squared shank bit shall be gripped in the socket, collet or jaw. The outer end of the bit shall be clamped in a bench vise with the body of the rod in a horizontal plane. The opposite end of the rod shall be supported and a torque of 80 inch-pounds applied to the square taper shank of the bit extension. The tapered shank of the bit shall not slip or turn in the socket, collet or jaw of the bit extension under this test. The bit shall then be released from the socket, collet or jaw of the bit extension without the aid of tools. Not failure, perceptible deformation or damage to the socket or collet shall have occurred as a result of this test (see 3.8).

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the requirements of PPP-P-40.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style, as applicable, of braces and bit extensions required (see 1.2.1).
(c) Sweep of braces required (see table I).
(d) Length of bit extensions required (see 3.8).
(e) Coating, if other than specified (see 3.3).

6.1.1 When repair parts are required, specify the following:

(a) The manufacturer's name or trademark of the brace for which the part is required.
(b) Quantity and the proper name or description of the part required.
(c) Type, class, style, and manufacturer's number of the brace for which the part is required (see 1.2).

MILITARY INTEREST:

Custodians:
Army - GL (MC4)
Navy - SH
Air Force - 84

User activities:
Army - ME, AV
Navy - YD

Preparing activities:
GSA-FSS

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